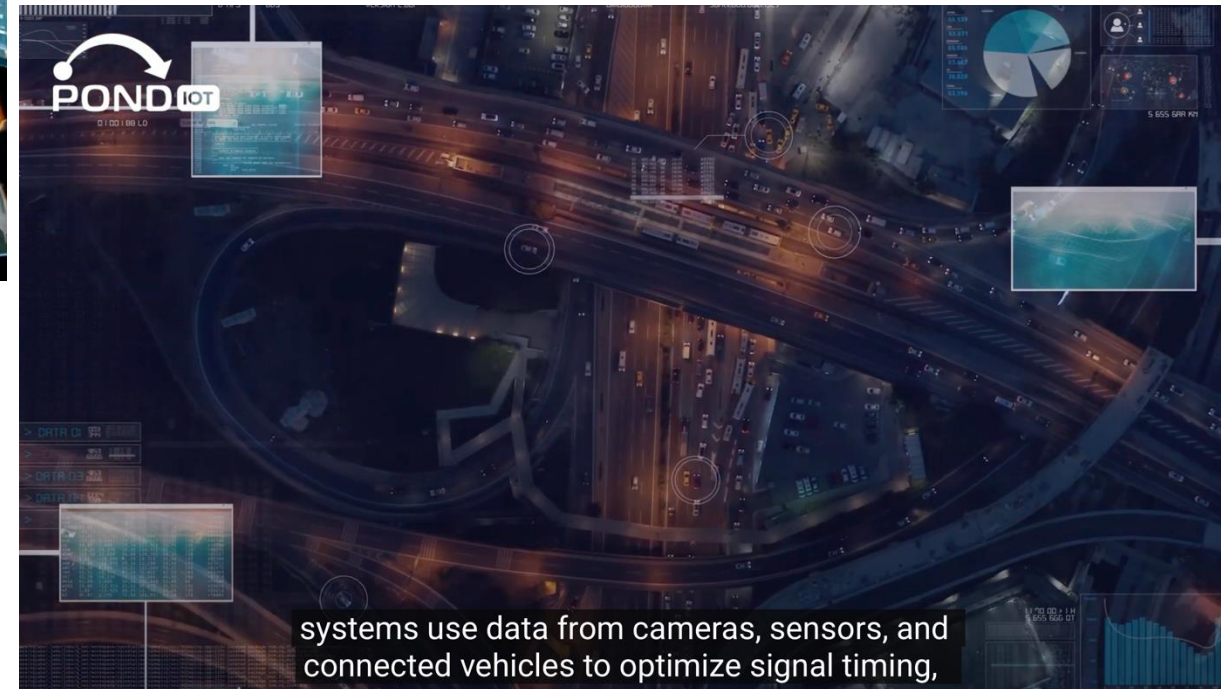




also benefits from CPS,



systems use data from cameras, sensors, and connected vehicles to optimize signal timing,

“The definitions of cyber-physical systems (CPS) and the Internet of Things (IoT) are converging over time to include a common emphasis on hybrid systems of interacting digital, analog, physical, and human components in systems engineered for function through integrated physics and logic.

CPS and IoT enable innovative applications [...] including in energy infrastructures, advanced manufacturing, building control, transportation, health care, and others.”

Cyber-Physical Systems and Internet of Things Foundations | NIST

<https://www.nist.gov/programs-projects/cyber-physical-systems-and-internet-things-foundations>



CPS/IOT Ecosystem: preparing Austria for the next digital revolution

<https://cpsiot.at/cps-iot-ecosystem-preparing-austria-for-the-next-digital-revolution/220/>

M.EIC043 / M.IA036

Sistemas Ciberfísicos e Internet das Coisas

Cyberphysical Systems and Internet of Things

Cyberphysical Systems and Internet of Things

This curricular unit **provides** comprehensive coverage of cyberphysical systems (CPS) and Internet of Things (IoT) technologies, emphasizing theoretical foundations, practical implementation skills, and critical evaluation capabilities.

The course **integrates** advanced technical knowledge with hands-on project experience to prepare students for research and industry roles in this rapidly evolving field.

The pedagogical approach **ensures** students develop both technical expertise and professional competencies essential for careers in cyber-physical systems and Internet of Things, while fostering innovation, critical thinking, and collaborative problem-solving capabilities.

Objectives

Technical

- Develop a comprehensive understanding of current trends, architectures, and computational models in CPS/IoT technologies
- Master practical skills in IoT technology platforms, wireless connectivity protocols, and edge computing solutions
- Acquire expertise in system integration methodologies, middleware platforms, and interoperability standards

Analytical and Design

- Cultivate critical thinking abilities to analyse trade-offs, evaluate design alternatives, and assess system performance characteristics
- Develop design competencies for creating integrated CPS/IoT systems that address real-world challenges while considering quality requirements
- Foster innovation capabilities to propose novel paradigms and solutions in emerging CPS and IoT applications

Professional and Research

- Build research competencies including literature review, methodology application, and systematic documentation of findings
- Enhance collaboration skills through multidisciplinary teamwork and effective communication with technical and non-technical stakeholders
- Prepare for professional practice by understanding ethical implications, security considerations, and societal impact of CPS/IoT technologies

Learning Outcomes and Competences

Learning outcomes are structured to progress from foundational knowledge to advanced analytical and creative capabilities.

Upon successful completion of this curricular unit, students will be able to:

LO1: Demonstrate comprehensive knowledge of CPS and IoT fundamentals

LO2: Comprehend architectural principles and computational models

LO3: Apply technical knowledge to system design challenges

LO4: Analyze system requirements and performance characteristics

LO5: Evaluate and critique system design solutions

LO6: Design and create integrated CPS/IoT systems

Knowledge and Understanding LOs

LO1: Demonstrate comprehensive knowledge of CPS and IoT fundamentals

Identify and define key concepts, terminologies, and components within cyber-physical systems and IoT architectures

Explain the evolution, current trends, and future directions of CPS and IoT technologies

Describe typical use cases across various domains and differentiate their specific system requirements

LO2: Comprehend architectural principles and computational models

Explain different architectural patterns, reference models, and design methodologies used in CPS and IoT systems

Discuss the theoretical foundations and justify the selection of appropriate computational models for specific applications

Compare various middleware platforms, communication protocols, and interoperability standards

Application and Analysis LOs

LO3: Apply technical knowledge to system design challenges

Implement IoT device connectivity using appropriate wireless technologies and communication protocols

Configure and deploy edge computing solutions and publish-subscribe communication models

Demonstrate proficiency in using contemporary development platforms and tools for CPS/IoT implementation

LO4: Analyze system requirements and performance characteristics

Analyze the trade-offs between different wireless connectivity options and their impact on system performance

Examine dependability, security, and privacy challenges and assess their implications for system design

Investigate and document existing CPS and IoT systems using systematic methodological approaches

Evaluation and Synthesis LOs

LO5: Evaluate and critique system design solutions

Evaluate alternative design approaches for CPS and IoT systems based on quality requirements and constraints

Critique existing implementations and justify design decisions using evidence-based reasoning

Assess the broader impact of CPS and IoT technologies on large-scale system development and societal implications

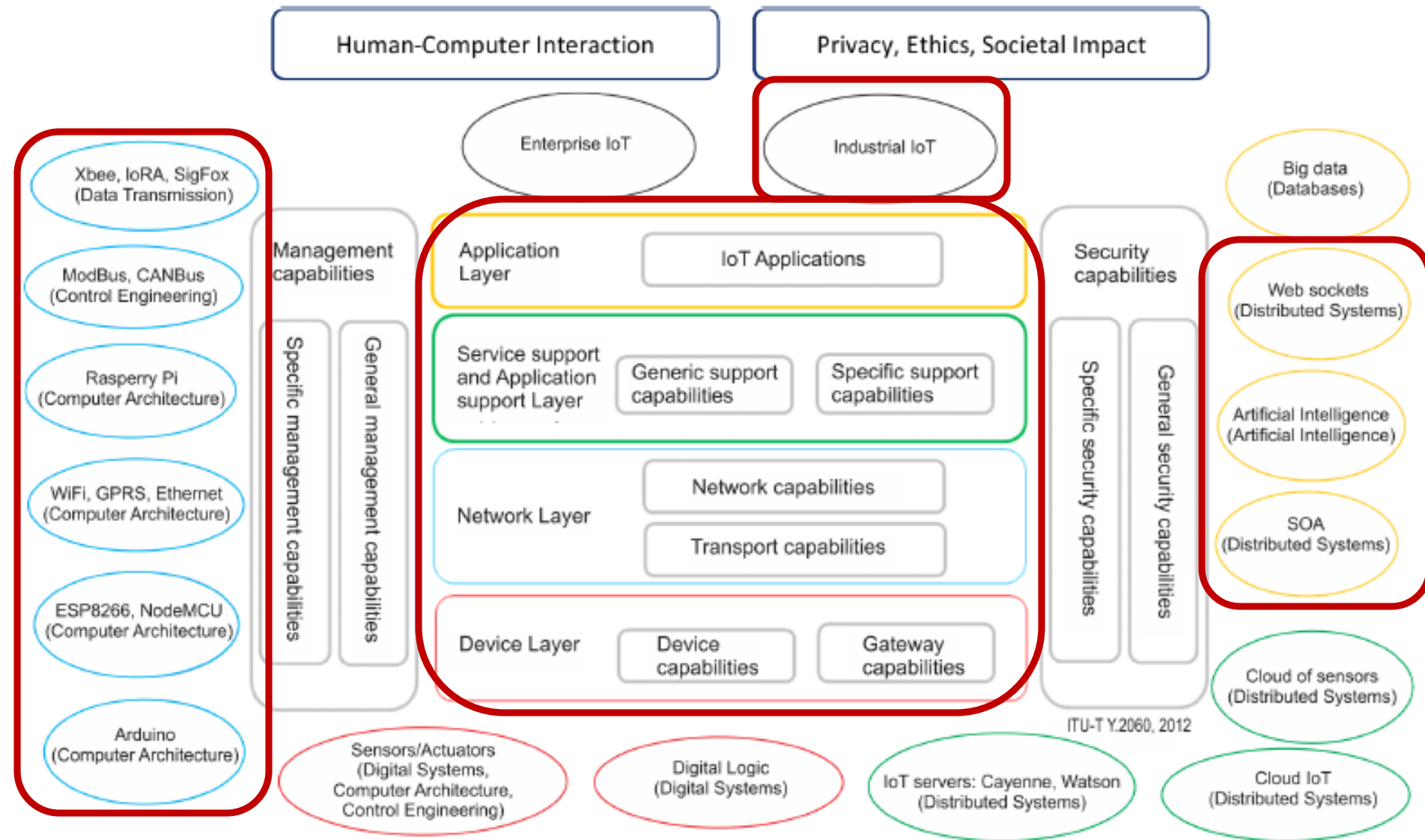
LO6: Design and create integrated CPS/IoT systems

Design and develop complete CPS and IoT-based systems by integrating multiple hardware and software components

Create innovative solutions that address real-world challenges while considering interdisciplinary constraints

Synthesize knowledge from multiple domains to propose novel paradigms and approaches in CPS and IoT

Topics Covered



Program

Module 1: CPS and IoT Foundations and Reference Models

Scope: Fundamental concepts, definitions, and use cases across various application domains

Key Topics: Reference architectures, system requirements analysis, emerging trends and paradigms

Learning Focus: Knowledge acquisition and conceptual understanding of the field

Module 2: System Architecture and Computational Models

Scope: Advanced architectural patterns and computational frameworks for CPS design

Key Topics: Networked embedded systems, models of computation, dependability principles, architectural views and patterns

Learning Focus: Understanding design principles and architectural decision-making

Program

Module 3: Edge Devices and Connectivity Technologies

Scope: Hardware platforms, communication protocols, and end-to-end connectivity solutions

Key Topics: Device constraints, wireless technologies, protocol performance analysis, edge computing paradigms

Learning Focus: Technical implementation and protocol selection strategies

Module 4: Middleware Platforms and Interoperability

Scope: Integration technologies and semantic interoperability frameworks

Key Topics: Publish-subscribe communication models, platform comparison, semantic web technologies, data integration

Learning Focus: System integration and platform evaluation methodologies

Program

Module 5: Dependability, Security, and Privacy

Scope: Quality attributes and risk management in CPS/IoT systems

Key Topics: Security threats, privacy preservation techniques, fault tolerance, trust management

Learning Focus: Critical evaluation of system vulnerabilities and mitigation strategies

Module 6: Novel Paradigms and Future Directions

Scope: Emerging technologies and research frontiers in CPS/IoT

Key Topics: Artificial intelligence integration, edge computing applications, sustainability considerations, ethical implications

Learning Focus: Innovation thinking and research orientation

Teaching Methods & Learning Activities

Lectures: introduce and consolidate the core theoretical concepts and background knowledge related to CPS/IoT.

Guest Expert Sessions: industry practitioners and researchers presenting current challenges, real-world implementations, and emerging trends.

Interactive Seminars: discussions on complex topics and case study analysis to promote collaborative learning and peer discussion and scientific reasoning.

Project-Based Active Learning (full-semester project) design, implement, and evaluate complete CPS/IoT systems:

- Definition of an industry-relevant problem in the CPS/IoT domain
- Identification of requirements and success criteria
- Literature review and state-of-the-art analysis
- PoC design, implementation, and experimental evaluation
- Analysis and reflection on functional and non-functional quality aspects (including dependability, security, and performance criteria)
- Students are encouraged to use real hardware when possible and to align the work with research on industry-relevant problems.

Course plan and (tentative) timeline

Deliverable			Initial project presentation* (topic and plan)			mini-exam**			Intermediate project presentation*** (architecture and PoC)			mini-exam**				Final project presentation****	Project final report	
Week	1	2	3	4	5	6		7	8	9	10	11	12	13	-	-	14	15
Date	16-9-2025	23-9-2025	30-9-2025	7-10-2025	14-10-2025	21-10-2025	28-10-2025	4-11-2025	11-11-2025	18-11-2025	25-11-2025	2-12-2025	9-12-2025	16-12-2025	23-12-2025	30-12-2025	6-1-2026	13-1-2026
Session	Module 1	Module 2	Module 3	Module 4		Module 5	Academic Week (semana da FEUP)		Interactive seminar	Interactive seminar	Module 6	Guest expert session	Guest expert session	Guest expert session				

6 lectures (complemented with on-line material) + 3 industry/research talks + 2 seminars
full-semester project: design and develop a CPS/IoT solution

* 5 minutes presentation

** open book / 45 minutes

*** 15 minutes presentation

**** 30 minutes presentation & demo

Assessment

Continuous assessment

2 intermediate individual assessments (end of October and early December)

Full-semester project

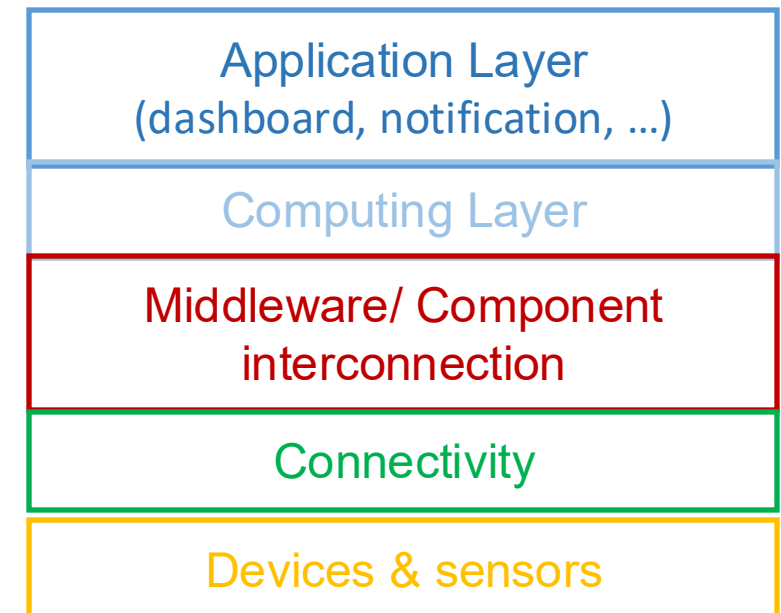
3 presentations (end of September, early October and January)

final report (mid January) (presentation slide deck + scientific paper)

Final grade: 30% continuous assessment + 70% full-semester project

Project Main Goals

- Design and implement a Proof of Concept (PoC) of a CPS/IoT solution to address an industry-relevant challenge (e.g. Industrial IoT, Inventory management/ supply chains, Monitoring industrial facilities and machinery, ..)
- Work in teams of 6/7 students.
- Platforms and devices (challenge independent for the PoC)
- Use of GitHub and keep a logbook (with weekly updates)



Project Hardware

Design, plan and implement an CPS/IoT application

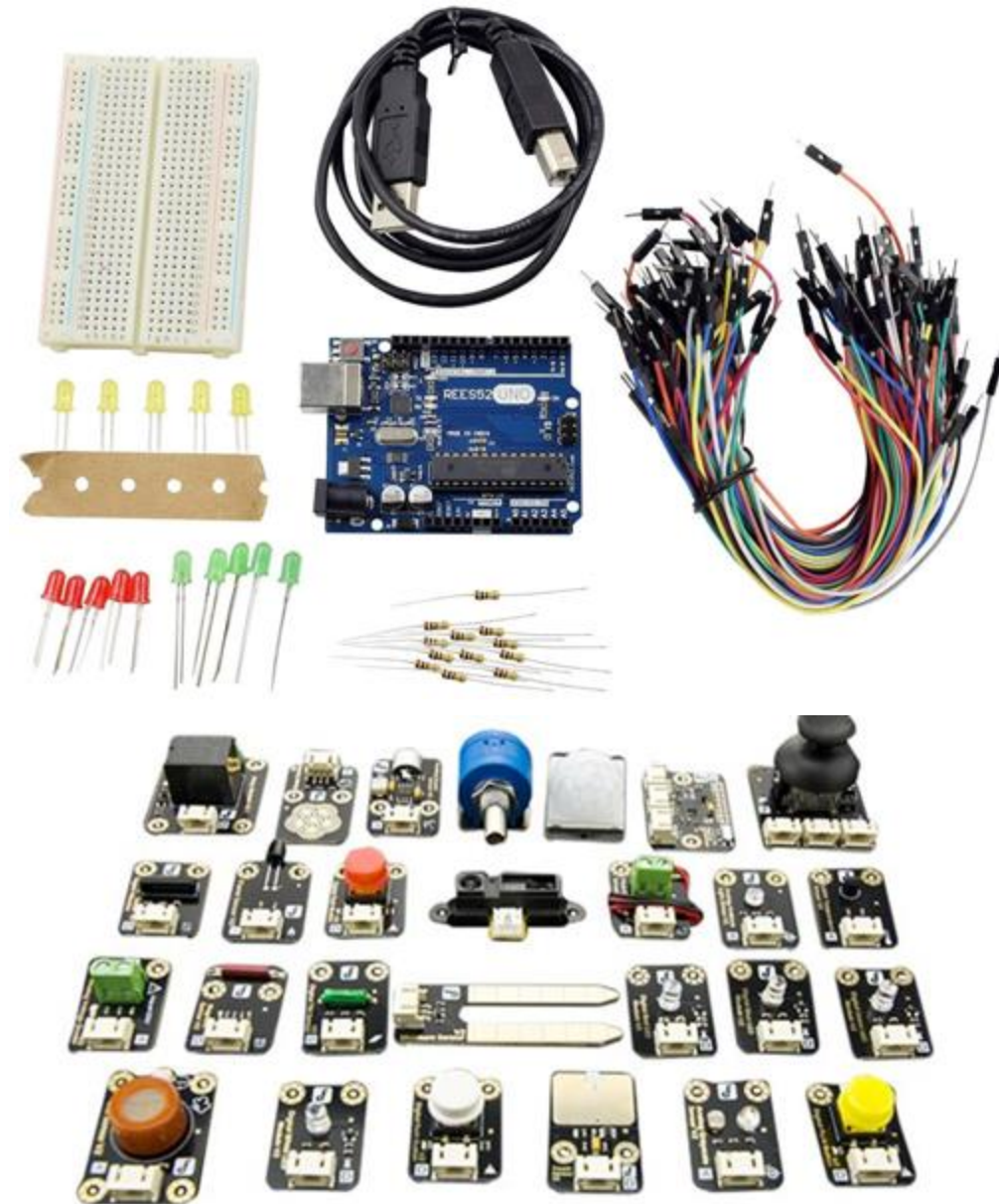
Available hardware

- Arduino NANO ESP32
- Raspberry Pi
- Sensors and actuators
- other

(List of required material by (or before) the intermediate presentation)

Access to workshop facilities (including 3D) for prototyping the PoC

Focus on complete CPS/IoT systems addressing industry-relevant challenges





Project Ideas

- I. **Smart Factory Predictive Maintenance:** Deploy sensors (e.g. vibration, temperature, current, pressure, ..) on industrial machinery to collect operational data and apply machine learning for failure prediction. Focus includes sensor integration, data acquisition, analytics, edge/cloud data transfer, and actuated alerts for maintenance scheduling.
- II. **Remote Industrial Process Monitoring:** Build a system that collects, visualizes, and controls industrial process data (e.g., tank levels, flow rates) using IoT-enabled devices or microcontrollers. Emphasize interoperability (e.g., Modbus/MQTT), real-time feedback, security, and HMI (human-machine interface) integration.
- III. **Smart Energy Management for Industrial Plants:** Design IoT-based power and energy consumption monitoring, with real-time optimization using control algorithms. Integrate sensors/meters, edge computing, and cloud dashboards for visualization and reporting.
- IV. **Digital Twin Prototype for Industrial Equipment:** Develop a digital replica using real-time sensor data to monitor, analyze, and forecast physical equipment states and maintenance needs. Emphasize integration of analytics, visualization, and feedback for decision support.
- V. **Environmental Monitoring for Industrial Sites:** IoT-based system for monitoring air/water quality, emissions, or hazardous gas levels within industrial facilities. Apply multi-sensor fusion, alerting logic, and reporting for regulatory compliance.
- VI. **Smart Camera for Industrial Sites:** IoT-based camera system for tracking assets in industrial facilities, analysing image feeds at the edge and managing events. Emphasize privacy, managing events and feedback for decision support.
- VII. **Industrial Cyber Physical Testbed:** Construct a networked industrial control system simulated environment to test CPS architectures. Cover testing, protocol, performance and uses.

VIII....

Projects from Previous Editions

- ❑ **Inventory Management (22/23):** [Slide Deck](#)
- ❑ **Environmental Monitoring for Smart Cities (22/23):** [Slide Deck](#)
- ❑ **Fleet Management in Industrial IoT (23/24):** [Slide Deck](#)
- ❑ **Smart Farming (23/24):** [Slide Deck](#)
- ❑ **IoT-based Smart Parking (23/24):** [Slide Deck](#)
- ❑ **SmartPool: Internet of Tanks (24/25):** [Demo Video](#) | [Scientific Paper](#)
- ❑ **SmartTank (24/25):** [Demo Video](#) | [Scientific Paper](#)
- ❑ **FurSight (24/25):** [Demo Video](#) | [Scientific Paper](#)

